

How to Enable Boot from QSPI Flash

1. Introduction

The i.MX RT Series is industry's first crossover processor provided by NXP. This document describes how to program a bootable image into the external storage device. For Information about Flashloader, MfgTool, please refer to the application note [“How to Enable Boot from HyperFlash and SD Card”AN12107](#).

The software used for example in this document are based on the i.MXRT1050 SDK. The development environment is IAR Embedded Workbench 8.20.1. The hardware development environment is MIMXRT1050-EVK Board. The version of flashloader is V1.0.

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2. MIMXRT1050 EVK board settings

2.1. EVKA Settings

In order to enable the onboard QSPI Flash features, EVK board (EVKA Board) settings need to be changed.

Step1:

- The onboard HyperFlash should be removed, otherwise it will impact the QSPI Flash read and write timing.

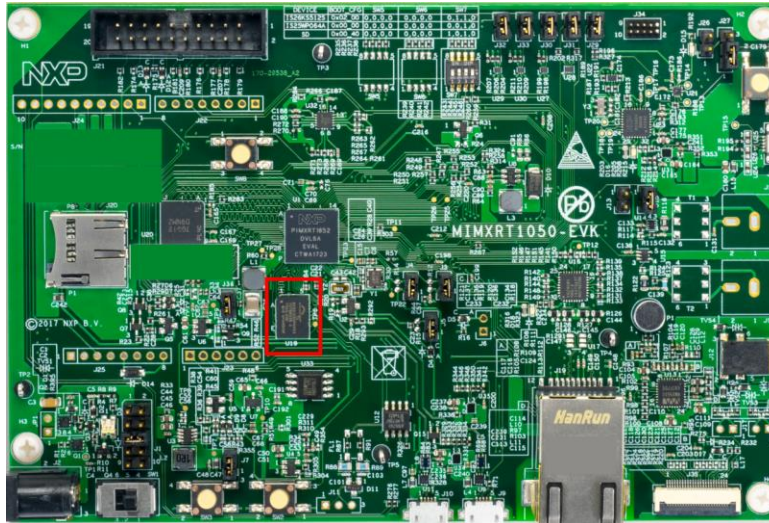


Figure 1. Remove the HyperFlash

Step2:

- Weld 0 Ω resistor to the pad from R153 to R158.

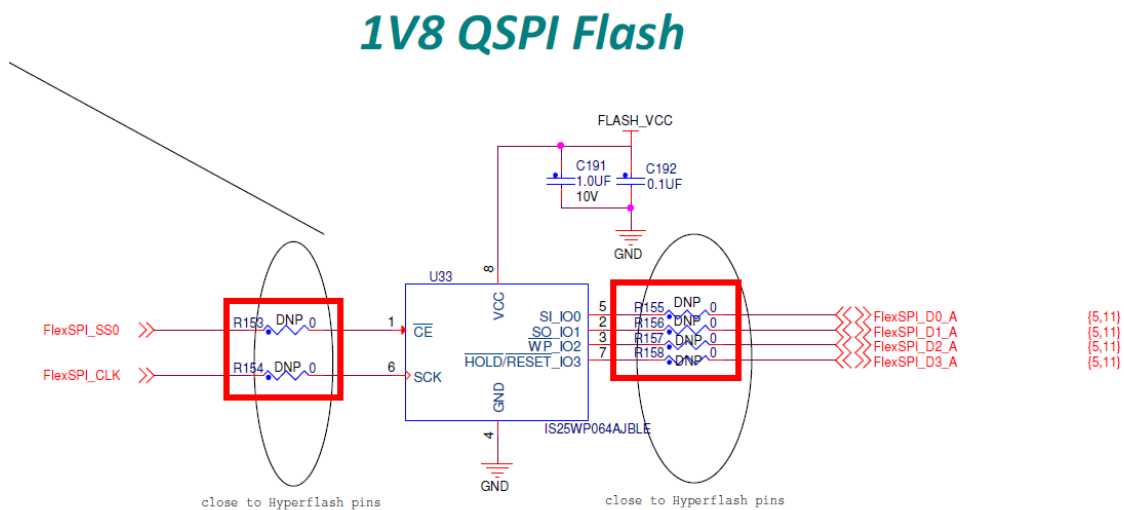


Figure 2. Weld 0 Ω resistor to the pad from R153 to R158

Step3:

- The firmware of OpenSDA needs to be replaced. The default firmware onboard is used to HyperFlash, so that the firmware should be replaced to QSPI Flash. Both HyperFlash and QSPI Flash's firmware can be downloaded from [NXP Website](#).

2.2. EVKB Settings

For EVKB board, the onboard HyperFlash doesn't need to remove.

Removed resistors: R356, R361 - R366.

Weld 0 Ω resistors: R153 - R158.

Follow the Step3 of Section 2.1 to update the OpenSDA firmware.

After those steps, the onboard QSPI Flash is ready to use.

NOTE

Even if QSPI flash itself doesn't have DQS pin, please keep it to be floating and enable it to get a higher read/write frequency. Please refer to table 35 and table 36 from [RT1050 datasheet](#), if DQS pin is not used, only 60MHz frequency of operation is supported while could up to 133MHz frequency of operation if DQS pin enabled for input timing.

4.5.2.1.1 SDR mode with FlexSPIn_MCR0[RXCLKSRC] = 0x0, 0x1

Table 35. FlexSPI input timing in SDR mode where FlexSPIn_MCR0[RXCLKSRC] = 0X0

Symbol	Parameter	Min	Max	Unit
	Frequency of operation	—	60	MHz
T _{IS}	Setup time for incoming data	8.67	—	ns
T _{IH}	Hold time for incoming data	0	—	ns

Table 36. FlexSPI input timing in SDR mode where FlexSPIn_MCR0[RXCLKSRC] = 0X1

Symbol	Parameter	Min	Max	Unit
	Frequency of operation	—	133	MHz
T _{IS}	Setup time for incoming data	2	—	ns
T _{IH}	Hold time for incoming data	1	—	ns

Figure 3. SDR mode input timing parameter

3. Program tools

3.1. DAP-Link (OpenSDA MSD drag/drop)

- QSPI Flash on EVK only.
- Binary file supports only.

NOTE

The default firmware of DAP-Link on EVK supports HyperFlash only. The firmware of DAP-Link should be replaced if the QSPI flash drag/drop is used. The firmware can be downloaded from [NXP Web](#).

3.2. MFG tool

The MfgTool supports I.MXRT BootROM and KBOOT based Flashloader, it can be used in factory production environment. The Mfgtool can detect the presence of BootROM devices connected to PC and invokes “blhost” to program the image on target memory devices connected to i.MX MCU device.

The blhost is a command-line host program used to interface with devices running KBOOT based Bootloader, part of MfgTool release. sb file support only.

4. Examples

4.1. OpenSDA Drag/Drop and boot from QSPI Flash

This chapter describes the steps needed that program an image to QSPI Flash by using OpenSDA Drag/Drop. The steps are as follows:

Step1:

- Open the Hello world demo in the SDK and select the project configuration as flexspi_nor_debug.([Figure 4](#))

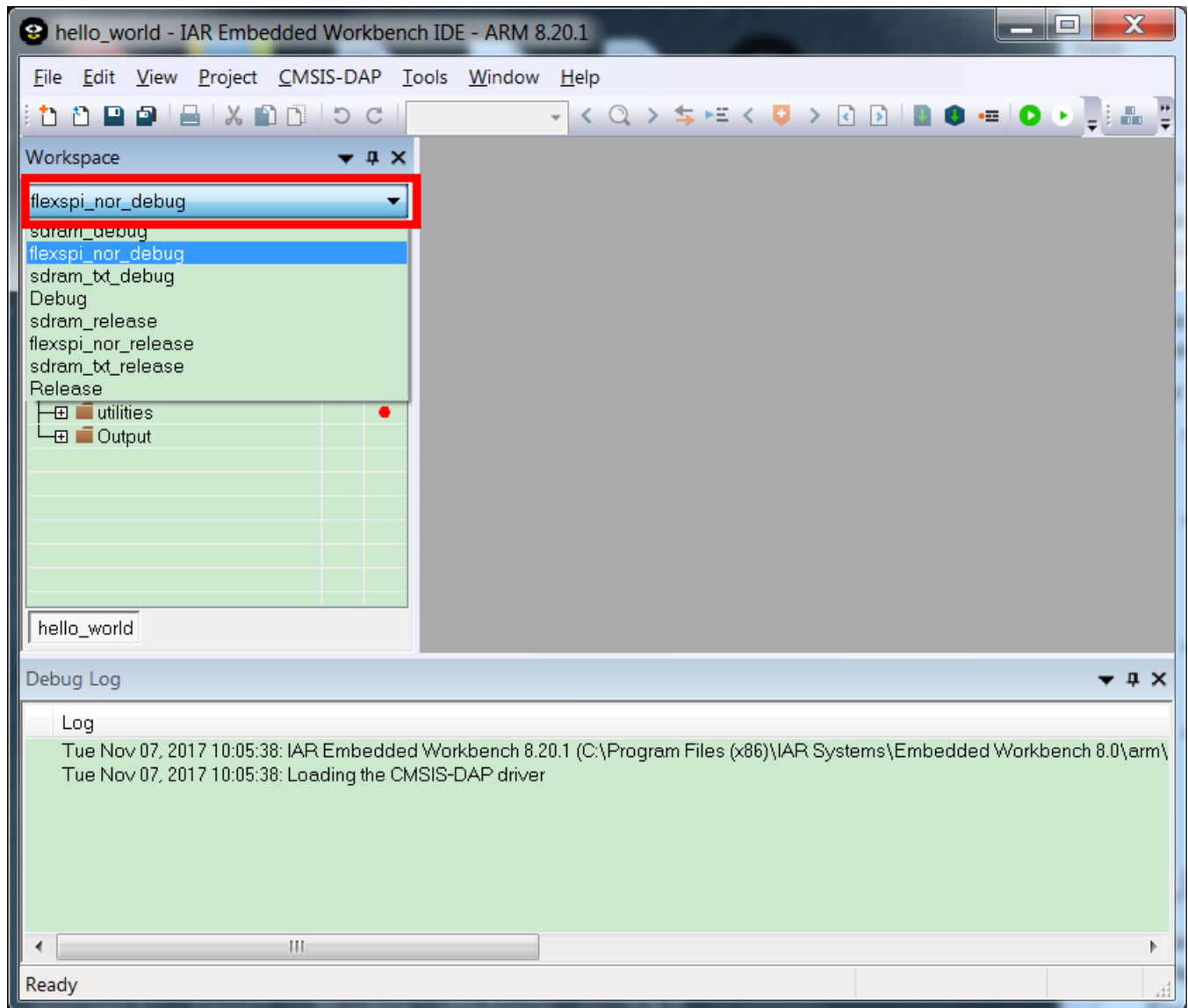


Figure 4. Select the project configuration as flexspi_nor_debug

Step2:

- Build the project and generate the image. You can find the hello_world.bin at following location ([Figure 5](#)):

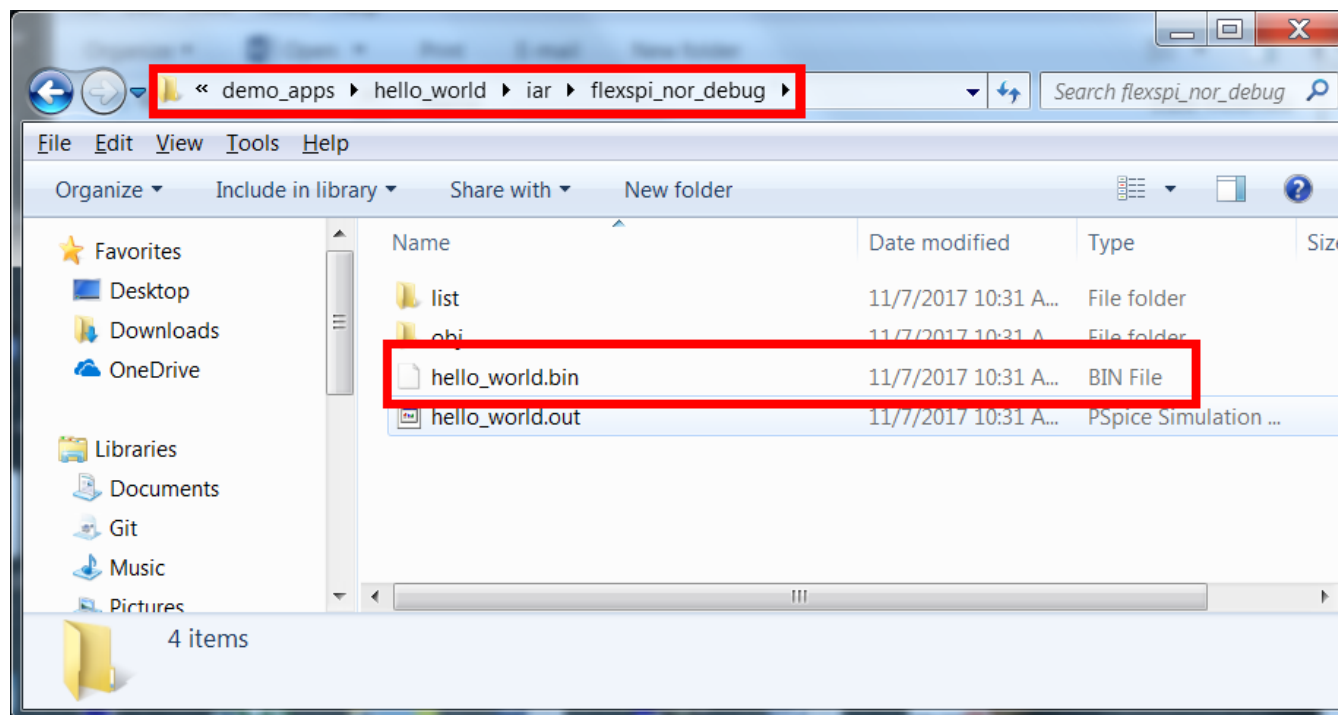


Figure 5. hello_world.bin location

Step3:

- Configure the board to serial downloader mode and make sure the power supply is from the Debug USB. To achieve these, SW7-4 should pull-up others pull-down [Figure 6](#) and the J1-5, J1-6 should be connected [Figure 7](#).

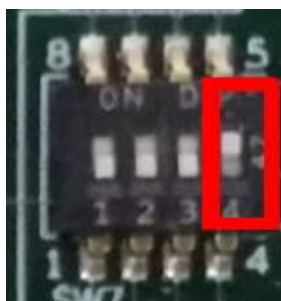


Figure 6. SW7-4 pull-up and others pull-down

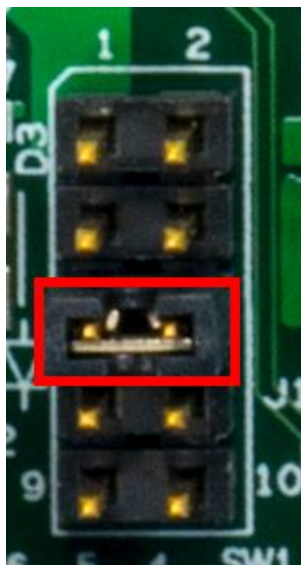


Figure 7. Power supply switch

Step4:

- Power up the board by connecting USB Debug Cable to J28 and open windows explorer and confirm that a U-Disk appears as a drive like [Figure 8](#).

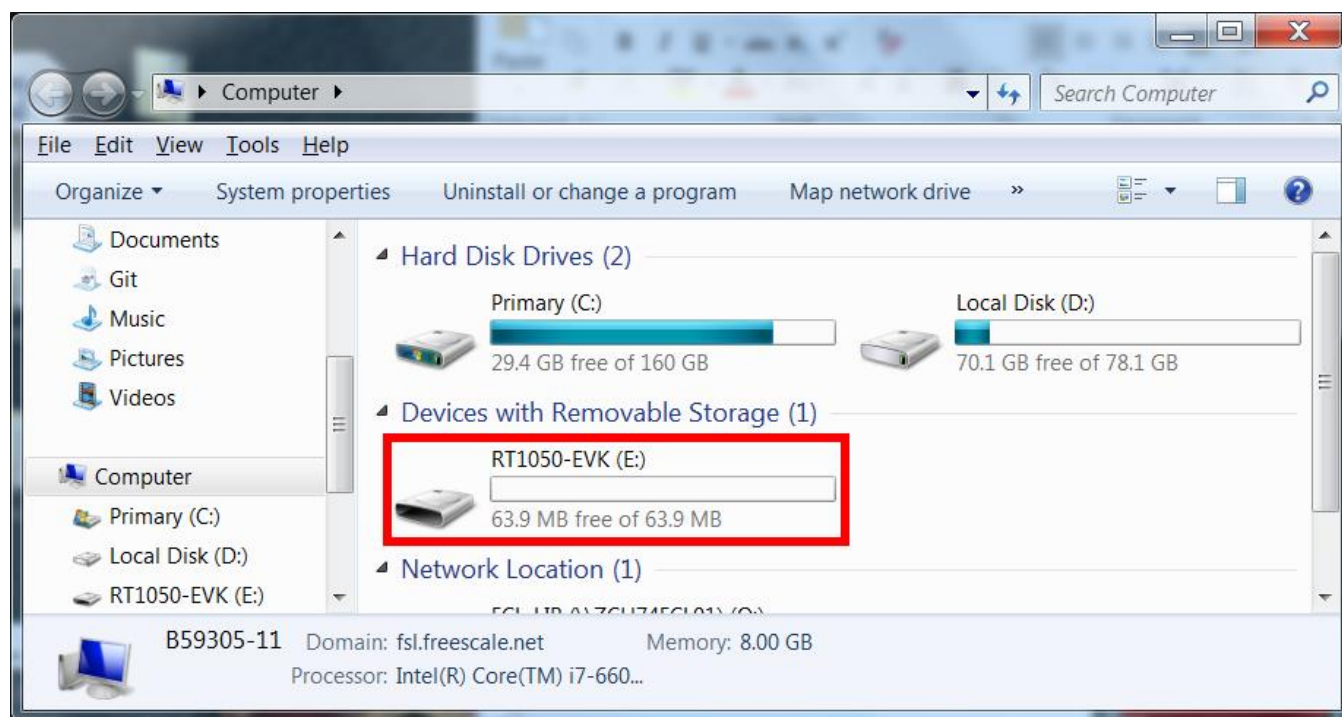


Figure 8. RT1050-EVK appeared

NOTE

The first time you connect the MBED USB to Host Computer Windows will ask to install the MBED serial driver.

Step5:

- Drag/Drop the hello_world.bin to RT1050-EVK. Then the RT1050-EVK disappears and after few seconds it will appear again.

Step6:

- Disconnect the USB Debug Cable, and configure the board to QSPI Flash Boot Mode which means SW7-3 pull-up others pull-down Figure 9.

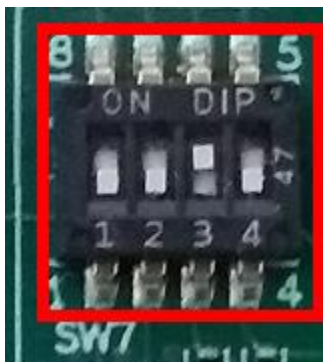


Figure 9. QSPI Flash Boot Mode Configuration

Step7:

- Connect the USB Debug Cable again and configure the Terminal Window:
 - Baud rate: 115200
 - Data bits: 8
 - Stop bit: 1
 - Parity: None
 - Flow control: None

Step8:

- Press SW3 to reset the EVK Board and “hello world” will be printed to the terminal. Figure 10

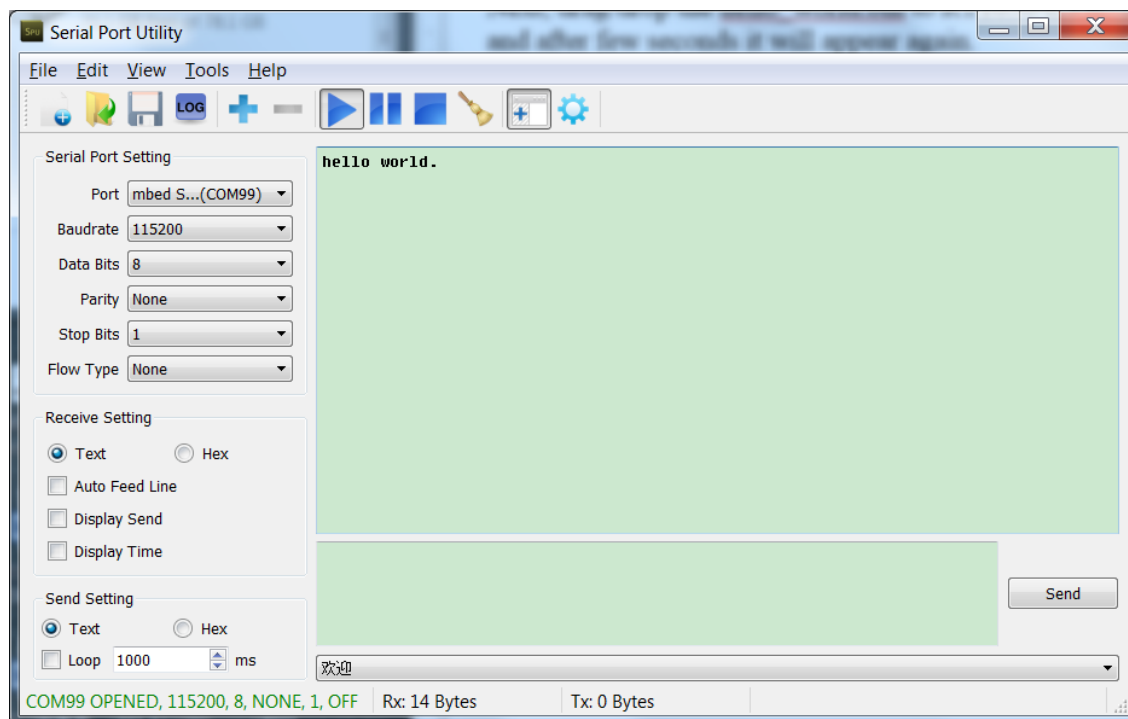


Figure 10. QSPI Flash Boot Mode Configuration

4.2. MFG Boot from QSPI Flash

This chapter describes the steps that using MFG tool to program an image to QSPI Flash and boot from the QSPI Flash.

Step 1:

- Open the Hello world demo in the SDK and select the project configuration as flexspi_nor_debug [Figure 11](#).

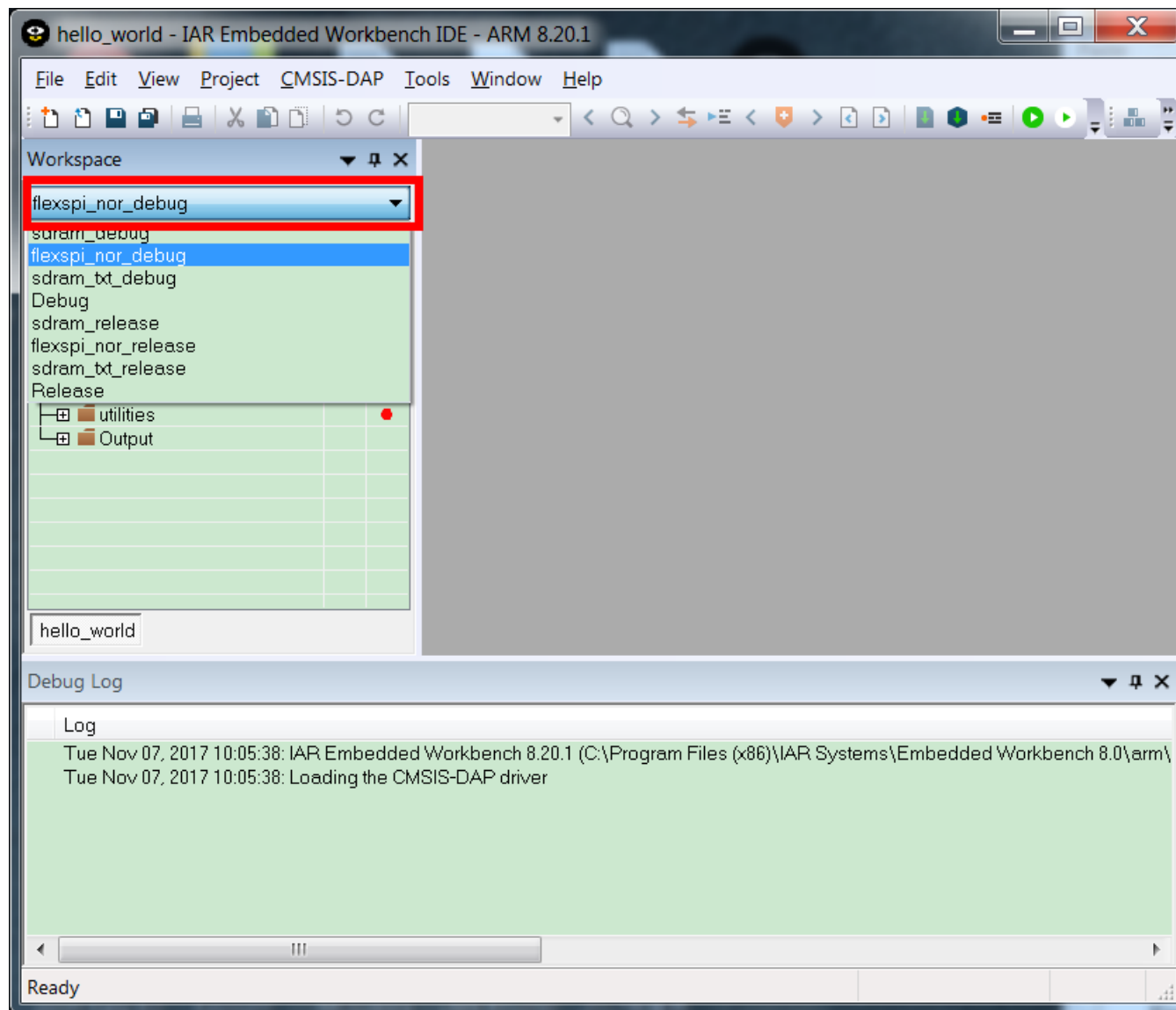


Figure 11. Select the project configuration as flexspi_nor_debug

Step 2:

- Build the project and generate the image. You can find the `hello_world.out` at following location [Figure 12](#):

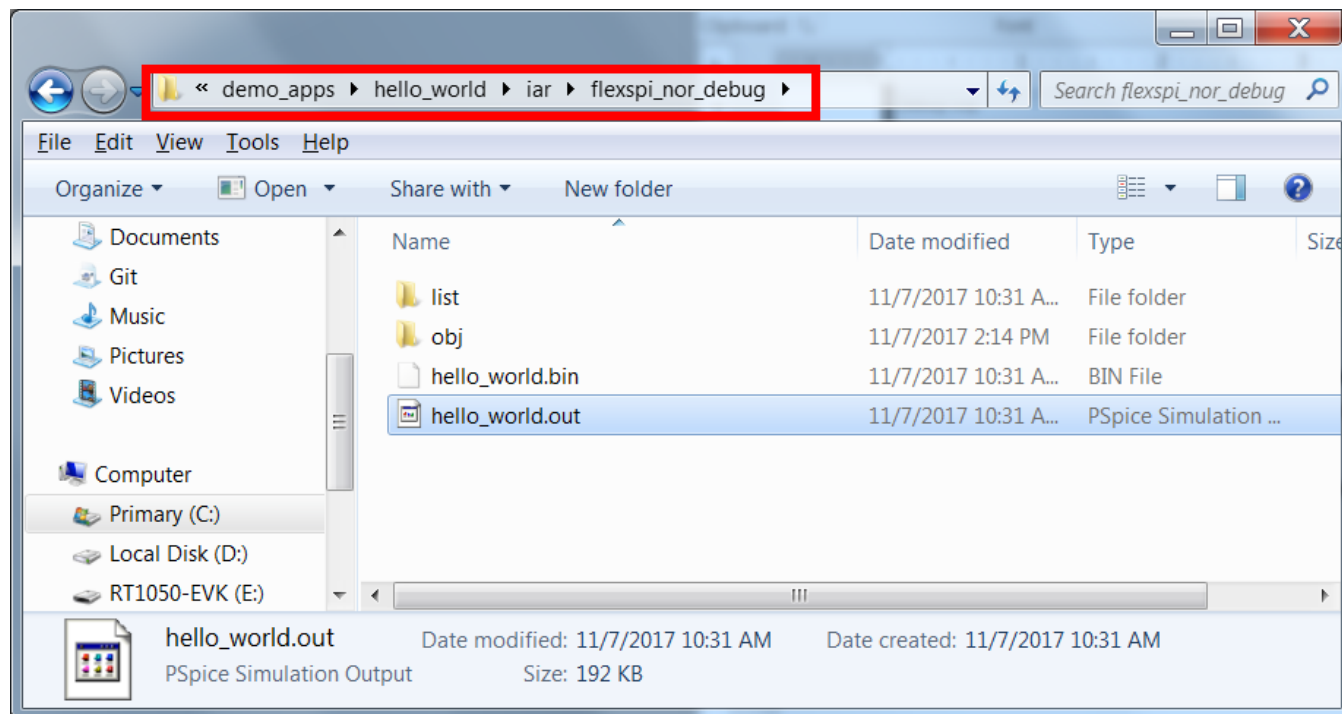


Figure 12. hello_world.out location

Step 3:

- Copy `hello_world.out` to the `elftosb` folder:

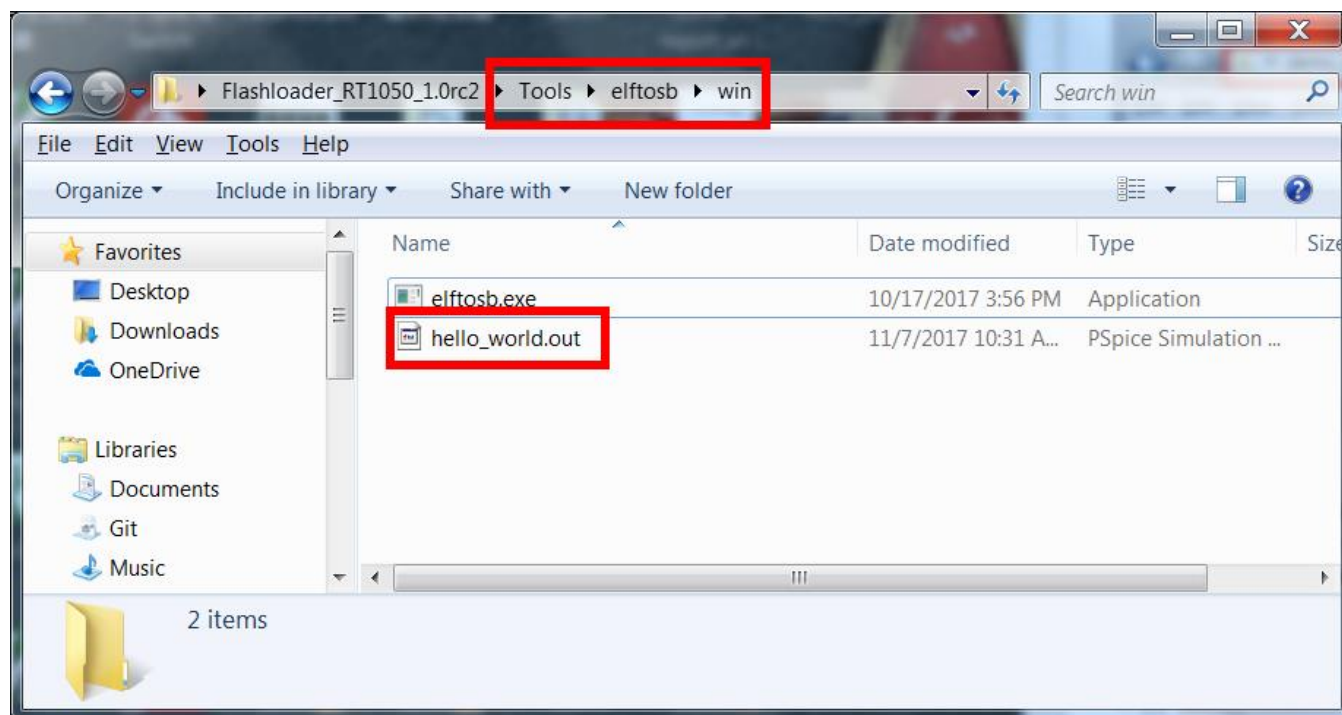


Figure 13. Copy hello_world.out

Step 4:

- Now we can use command to generate the i.MX Bootable image using elftosb file. Open cmd.exe and type following command:

```
elftosb.exe -f imx -V -c ../bd_file/imx10xx/imx-flexspinor-normal-unsigned.bd -o  
ivt_flexspi_nor_hello_world.bin hello_world.out
```

```

Administrator: C:\windows\system32\cmd.exe

C:\Users\b59305\Desktop\Flashloader_i.MXRT1050_1.0_FAE\Flashloader_RT1050_1.0\Tools\elftosb\win>elftosb.exe -f imx -U -c ../../bd_file/imx10xx/imx-flexspinor-normal-unsigned.bd -o ivt_flexspi_nor_hello_world.bin hello_world.out
Section: 0x0
iMX bootable image generated successfully

C:\Users\b59305\Desktop\Flashloader_i.MXRT1050_1.0_FAE\Flashloader_RT1050_1.0\Tools\elftosb\win>

```

Figure 14. Generate i.MX Bootable image

After above command, two bootable images are generated:

- ivt_flexspi_nor_hello_world.bin
- ivt_flexspi_nor_hello_world_nopadding.bin

ivt_flexspi_nor_hello_world.bin:

The memory regions from 0 to ivt_offset are filled with padding bytes (all 0x00s).

ivt_flexspi_nor_hello_world_nopadding.bin:

Starts from ivtdata directly without any padding before ivt.

The later one will be used to generate SB file for QSPI Flash programming in subsequent section.

Step 5:

This step creates a SB file for QSPI Flash programming. A boot_image.sb file will be generated that is for MfgTool use later. Because no bd file can be used for QSPI Flash, so that we need create a bd file which can be used for QSPI Flash.

- Copy the “program_flexspinor_image_hyperflash.bd” (located in Tools/bd_file/imx10xx) and named it as “program_flexspinor_image_qspinor.bd”. The new bd file will be used for QSPI

Flash. Open the bd file and modify parameter. Modify the parameter from 0xc0233007(Figure 15) to 0xc0000005.

```

31      #          3 - 8
32      # bit [15:12] CMD pads (Pads used for query Flash Parameters)
33      #          0 - 1
34      #          2 - 4
35      #          3 - 8
36      # bit [11: 08] fixed to 0
37      # bit [07: 04] fixed to 0
38      # bit [03: 00] Flash Frequency, device specific
39      #
40      # In this example, the 0xc0233007 represents:
41      #   HyperFLASH 1V8, Query pads: 8 pads, Cmd pads: 8 pads, Frequency: 133MHz
42      load 0xc0233007 > 0x2000;
43      # Configure HyperFLASH using option a address 0x2000
44      enable flexspinor 0x2000;
45
46      #2 Erase flash as needed.
47      erase 0x60000000..0x60010000;
48
49      #3. Program config block
50      # 0xf000000f is the tag to notify Flashloader to program FlexSPI NOR config block to th

```

Figure 15. Modify the parameter from 0xc0233007 to 0xc0000005

Open cmd.exe and type following command:

```

elftosb.exe -f kinetis -V -c ../../bd_file/imx10xx/program_flexspinor_image_qspinor.bd -o
boot_image.sb ivt_flexspi_nor_hello_world_nopadding.bin

```

```

Administrator: C:\windows\system32\cmd.exe

C:\Users\b59305\Desktop\Flashloader_i.MXRT1050_1.0_FAE\Flashloader_RT1050_1.0\Tools\elftosb\win>elftosb.exe -f imx -U -c ../../bd_file/imx10xx/imx-flexspino
rmal-unsigned.bd -o ivt_flexspi_nor_hello_world.bin hello_world.out
      Section: 0x0
iMX bootable image generated successfully

C:\Users\b59305\Desktop\Flashloader_i.MXRT1050_1.0_FAE\Flashloader_RT1050_1.0\Tools\elftosb\win>elftosb.exe -f kinetis -U -c ../../bd_file/imx10xx/program_flexs
pinor_image_qspinor.bd -o boot_image.sb ivt_flexspi_nor_hello_world_nopadding.bi
n
Boot Section 0x00000000:
  FILL | adr=0x00002000 | len=0x00000004 | ptn=0xc0000005
  ENA  | adr=0x00002000 | cnt=0x00000004 | flg=0x0900
  ERAS | adr=0x60000000 | cnt=0x00010000 | flg=0x0000
  FILL | adr=0x00003000 | len=0x00000004 | ptn=0xf000000f
  ENA  | adr=0x00003000 | cnt=0x00000004 | flg=0x0900
  LOAD | adr=0x60001000 | len=0x0000314e | crc=0x3ca3ee6c | flg=0x0000

C:\Users\b59305\Desktop\Flashloader_i.MXRT1050_1.0_FAE\Flashloader_RT1050_1.0\Tools\elftosb\win>

```

Figure 16. Create a SB file for QSPI Flash programming

After performing above command, the boot_image.sb is generated under elftosb folder Figure 17.

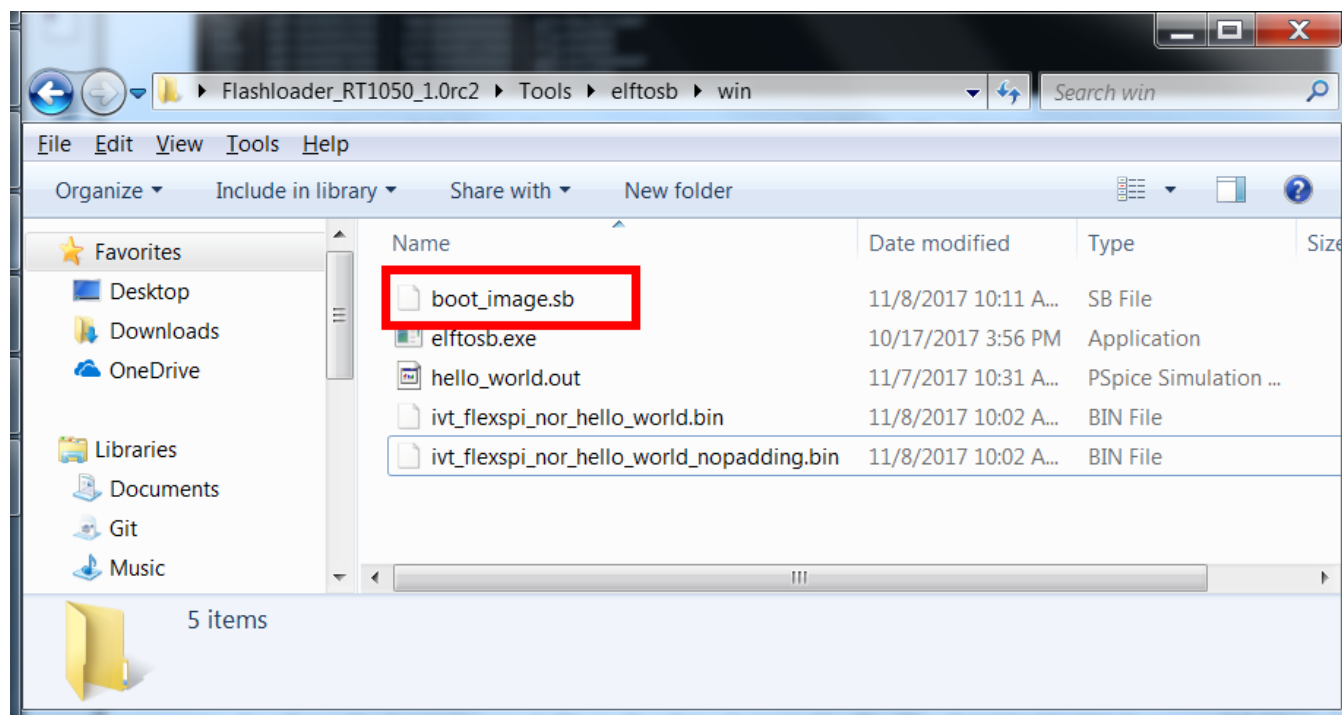


Figure 17. The boot_image.sb is generated

Step6:

- Copy the boot_image.sb file to OS Firmware folder:

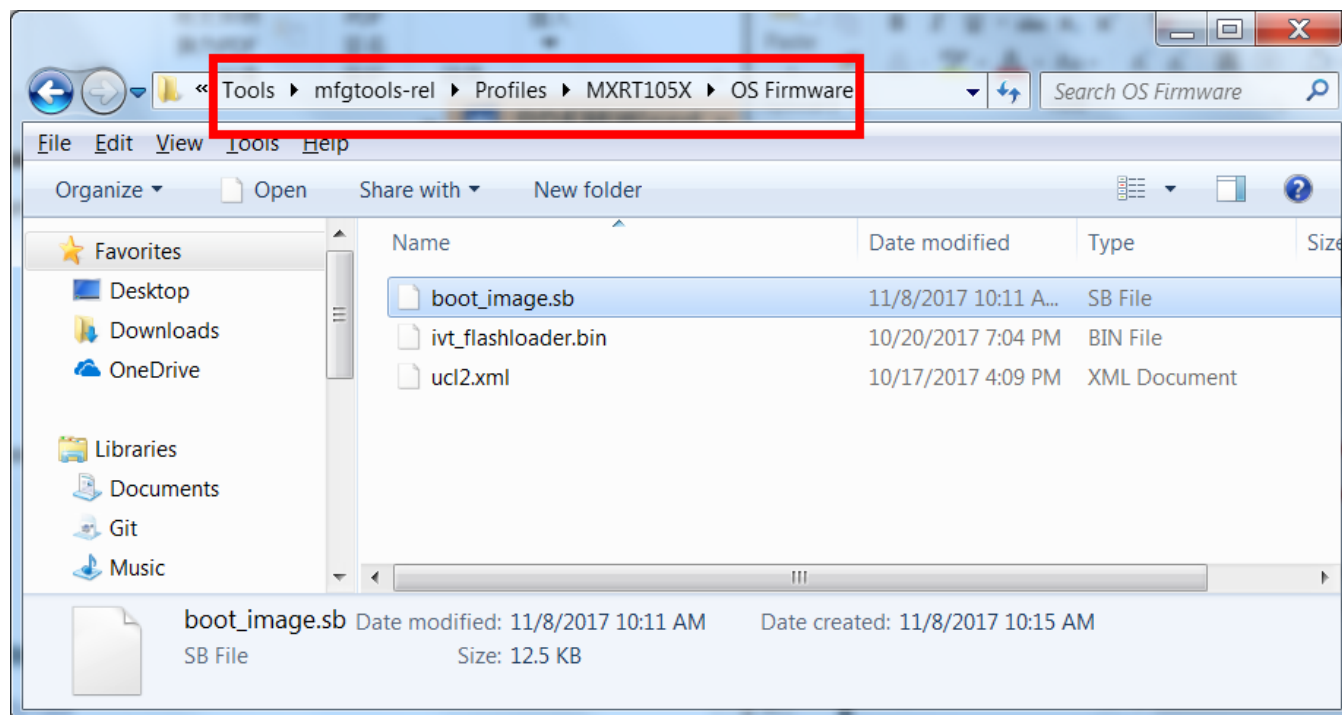


Figure 18. Copy the boot_image.sb to OS Firmware folder

Now, change the “name” under “[List]” to “MXRT105x-DevBoot” in cfg.ini file under <mfgtool_root_dir> folder.

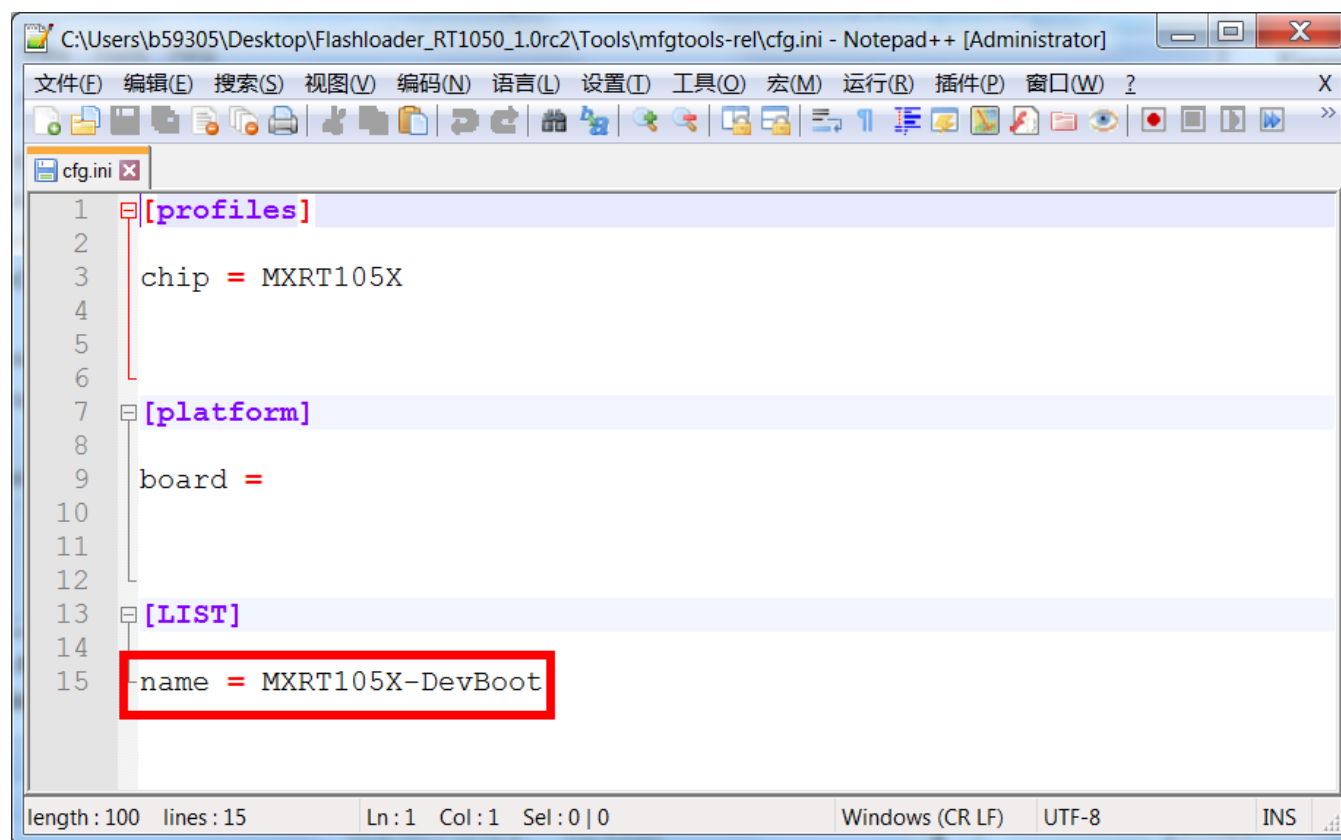


Figure 19. Change the name to “MXRT105x-DevBoot”

Switch the EVK-Board to Serial Downloader mode by setting SW7 to “1-OFF, 2-OFF, 3-OFF, 4-ON”. Then power up the EVK Board by inserting USB Cable to J9. Open MfgTool, it will show the detected device like Figure 20.

NOTE

In some corner case, HID-compliant device is not recognized which is because the PC only have USB root device and no USB hub device, and this software limitation will be fixed in near future, the workaround at this moment is to use external USB hub as extension.

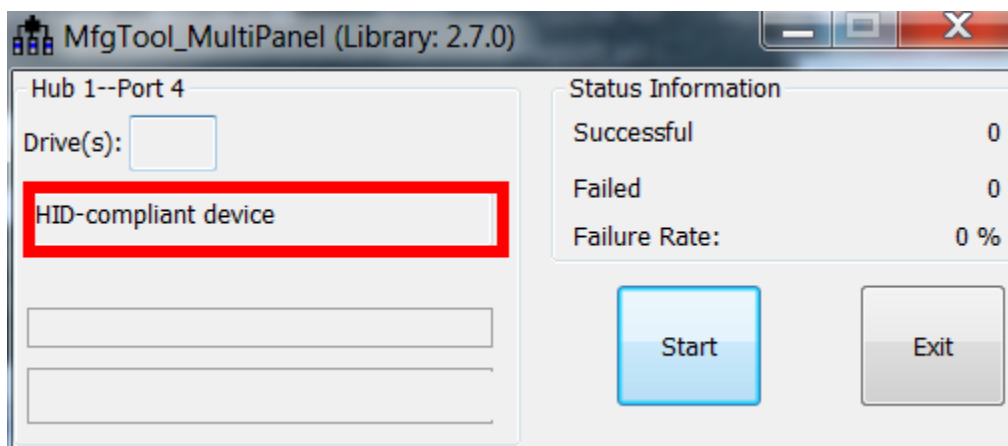


Figure 20. MfgTool GUI with device connected

Click “Start”, Mfgtool will do Mfgtool process and after all are done, MfgTool will show the success status as shown in Figure 21. Click “Stop” and Close the Mfgtool.

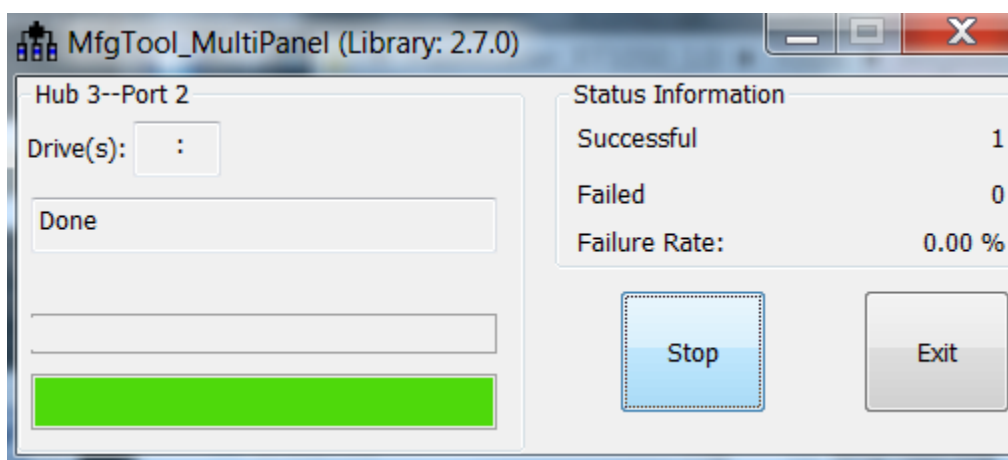


Figure 21. Successful Status

Step7:

- Switch the RT1050-EVK board to Internal boot mode and select QSPI Flash as boot device by setting SW7 to “1-OFF, 2-OFF, 3-ON, 4-OFF”. Connect the USB Cable to J28 and open a terminal, then reset the Board. We can see that “hello world” will be printed to the terminal.

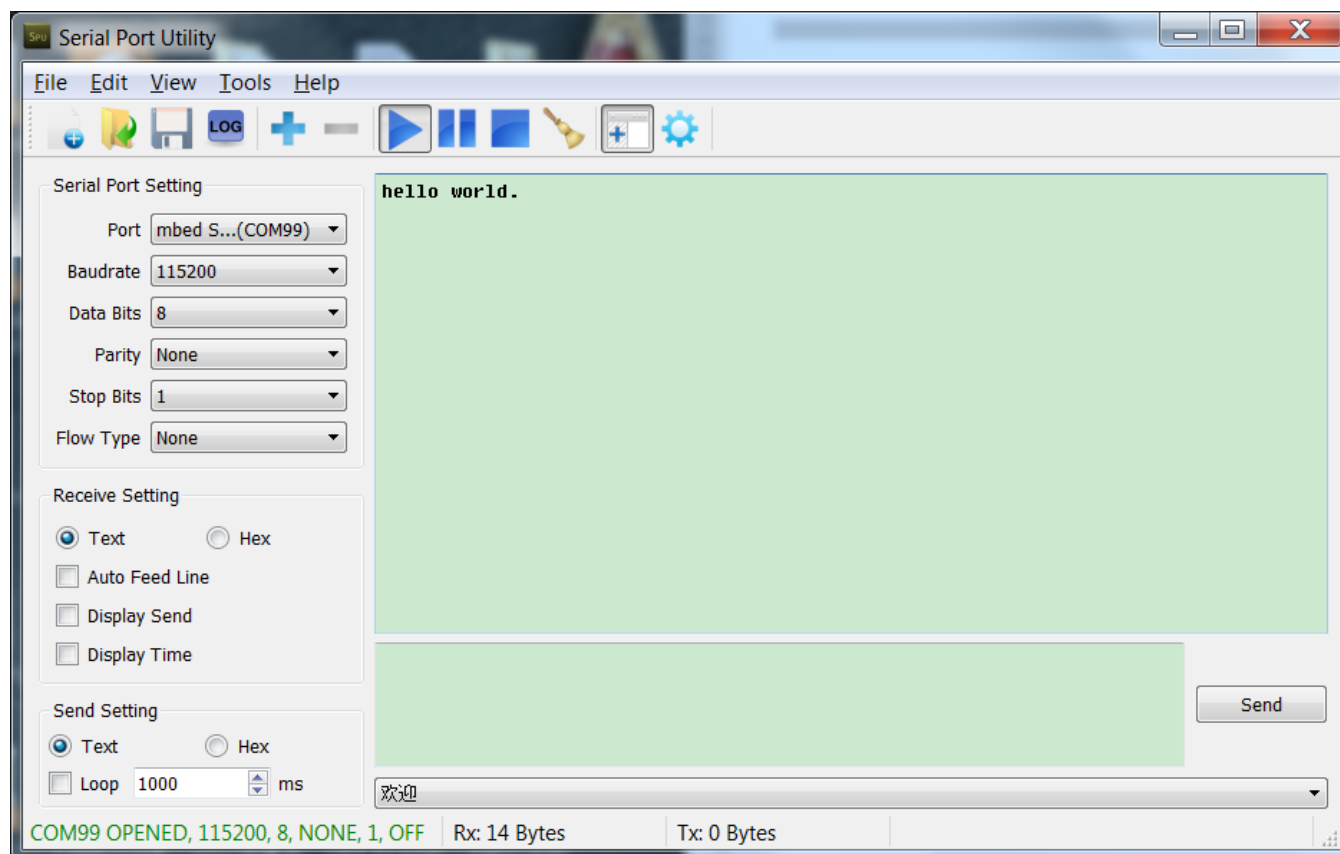


Figure 22. “hello world” be printed to the terminal

4.3. MFG Boot from QSPI Flash with DCD for SDRAM

For steps, please refer to [“How to Enable Boot from HyperFlash and SD Card”, AN12107](#).

5. QSPI Flash support list

Besides the EVK onboard QSPI Flash, the Flashes in [Table 1](#) are also supported and please note those are just typical examples with those flash vendors, theoretically we could support all the flash memory that comply with JESD216/JESD216A/JESD216B.

At the same time, the RT1050EVK could support both 1.8V and 3.3V SPI flash device by switching the FLASH_VCC power supply as below Figure 23 shows. If 3.3V SPI flash is mounted, you need mount R301 and DNP R49, otherwise you need mount R49 and DNP R301.

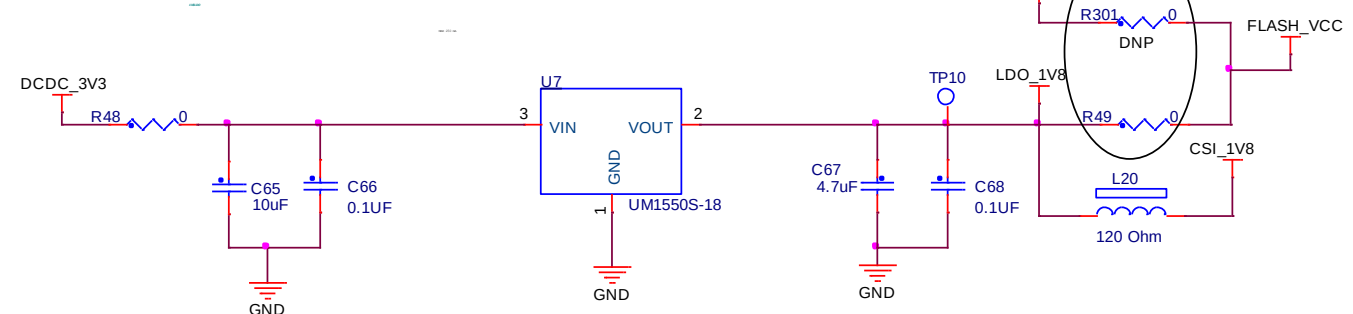


Figure 23. FLASH_VCC switching

Table 1. QSPI Flash support list

Vendor		Flash Part Number	Voltage
Macronix		MX25U6433F	3.3V
Macronix		MX25U6435E	1.8V
ISSI		IS25LP064A-JBLE	3.3V
ISSI		IS25WP064AJBLE	1.8V
GD		GD25Q64C	3.3V
GD		GD25LQ64C	1.8V
WINBOND		W25Q64JV	3.3V
WINBOND		W25Q64FW	1.8V
Micron		MT25QL128ABA1ESE-0SIT	3.3V
Micron		MT25OU128ABA1ESE-0SIT	1.8V

NOTE

Part of the list of flash need to use Flashloader v1.1.

6. Conclusion

This application note mainly describes how to use Flashloader step by step. For more information, Please refer to “[i.MX MCU Manufacturing User's Guide](#)” and “[How to Enable Boot from HyperFlash and SD Card](#)”.

7. Revision history

Table 2. Revision history

Revision number	Date	Substantive changes
0	02/2018	Initial release

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